Robots in the Open World: Autonomous Driving, Drones, Deliveries and Rescue

1. Short overview

Robots are no longer limited to factories and controlled environments. Thanks to advances in Artificial Intelligence (AI), sensors and computer vision, they can now operate in the real world, where they must interact with people, navigate unpredictable environments and make decisions independently. This is known as robotics in the open world. Some important applications include autonomous cars, drones used for different purposes, delivery robots that transport packages and food, and rescue robots that help emergency services during disasters. Although these technologies are becoming increasingly advanced, they still face challenges related to safety, unpredictable situations and reliability.

2. Fuller summary

The context: From controlled environments to the open world

In the past, most industrial robots operated in controlled environments, such as factories, where they performed repetitive tasks and followed predefined instructions. However, modern robotics aims to develop machines capable of operating in the real world, where conditions constantly change. Unlike factory robots, open-world robots must deal with unexpected obstacles, moving people, different weather conditions and unfamiliar environments. To achieve this, they use Artificial Intelligence, cameras, sensors, GPS and other technologies to understand their surroundings and make decisions autonomously. (Robotics in Unstructured Environments)

Autonomous Driving

One of the most important applications of open-world robotics is autonomous driving. Self-driving cars use cameras, LiDAR and radar to detect other vehicles, pedestrians, traffic lights and obstacles. Artificial Intelligence processes this information to understand the environment, predict what other road users might do and choose the safest route. One of the best-known companies in this field is Waymo, which developed autonomous vehicles capable of transporting passengers through real city streets without a human driver. These vehicles combine detailed maps with real-time sensor information to navigate traffic and react to unexpected situations. (Waymo)

Drones

Drones are another important example of robots operating in the open world. They can fly autonomously or with limited human supervision and are used for several purposes, including agriculture, infrastructure inspection, environmental monitoring, photography and search and rescue operations. Equipped with cameras, GPS and other sensors, drones can explore large or difficult-to-reach areas, collect information and identify potential problems. For example, during

natural disasters, drones equipped with thermal cameras can help emergency services locate missing people and assess dangerous areas without putting rescuers at risk. (EASA)

Autonomous Deliveries

Robots are increasingly being used to deliver food, packages and medical supplies. There are two main types: ground delivery robots, which move along sidewalks, and aerial delivery drones, which transport packages through the air. Starship Technologies, for example, developed small autonomous robots that use cameras, GPS and sensors to navigate sidewalks, avoid obstacles and deliver food and groceries. Another important example is Zipline, a company that uses autonomous drones to deliver medical products. In 2016, Rwanda launched the world's first national drone delivery service with Zipline, initially focusing on transporting blood to hospitals, including facilities in remote areas. This technology significantly reduces delivery times and helps healthcare services reach places that are difficult to access by road. (Starship Technologies, Zipline, Gavi)

Search and Rescue Robots

Robots can also play an important role during emergencies and natural disasters. Search and rescue robots are designed to enter dangerous environments, such as collapsed buildings, areas affected by earthquakes, fires and industrial accidents. They can be equipped with cameras, thermal sensors and other instruments to detect survivors, explore unstable structures and provide information to rescue teams. Both ground robots and aerial drones can be used in these situations. Their main advantage is that they can enter areas that may be too dangerous for human rescuers, reducing risks and helping emergency services understand the situation before entering. The National Institute of Standards and Technology (NIST) works on testing and developing performance standards for emergency response robots, focusing on their mobility, navigation, sensing and reliability. (NIST)

The challenges and the future

Despite their progress, robots operating in the open world still face several challenges. Unlike controlled environments, the real world is unpredictable: weather conditions can change, sensors may fail, GPS signals can be lost and unexpected obstacles can appear. Robots must also understand human behaviour and make safe decisions in complex situations. Another important issue is reliability: a robot used for medical deliveries or rescue operations must work correctly because mistakes can have serious consequences. In the future, improvements in Artificial Intelligence, machine learning and sensor technology could make robots more autonomous, adaptable and capable of performing increasingly complex tasks. However, ensuring their safety and reliability will remain essential. (Robotics in Unstructured Environments, NIST)

3. Short overview (Key words)

- Robots -> no longer limited to factories
- Open world -> real, unpredictable environments

- AI + sensors + computer vision -> autonomous decisions
- Autonomous driving -> self-driving cars
- Drones -> agriculture, inspection, monitoring, rescue
- Autonomous deliveries -> food, packages, medical supplies
- Search and Rescue -> earthquakes, fires, collapsed buildings
- Main challenges -> safety, reliability, unpredictable situations
- Future -> more autonomous and adaptable robots

4. Fuller summary (Key words)

The context: From controlled environments to the open world

- Past -> robots mainly used in factories
- Controlled environments -> repetitive tasks, predefined instructions
- Modern robotics -> operating in the real world
- Open world -> unpredictable environments
- Unexpected obstacles, people, weather conditions
- AI, cameras, sensors, GPS -> understand surroundings and make decisions

Autonomous Driving

- Self-driving cars
- Cameras + LiDAR + radar
- Detect vehicles, pedestrians, traffic lights and obstacles
- AI -> understand environment and predict movements
- Waymo -> autonomous vehicles
- Real city streets -> no human driver
- Detailed maps + real-time sensor information
- Navigate traffic and react to unexpected situations

Drones

- Autonomous or limited human supervision
- Applications:
    - Agriculture
    - Infrastructure inspection
    - Environmental monitoring
    - Photography
    - Search and rescue
- Cameras + GPS + sensors
- Explore large and difficult-to-reach areas
- Thermal cameras -> locate missing people
- Disaster response -> assess dangerous areas without risking rescuers

Autonomous Deliveries

- Ground robots -> sidewalks
- Aerial drones -> transport through the air
- Food, packages and medical supplies
- Starship Technologies:

    - Small autonomous delivery robots
    - Cameras + GPS + sensors
    - Navigate sidewalks and avoid obstacles
    - Deliver food and groceries
- Zipline:
    - Autonomous delivery drones
    - Rwanda - 2016
    - World's first national drone delivery service
    - Initially transported blood to hospitals
    - Remote areas -> difficult road access

Search and Rescue Robots

- Natural disasters and emergencies
- Earthquakes, fires, collapsed buildings, industrial accidents
- Ground robots + aerial drones
- Cameras + thermal sensors
- Detect survivors and explore dangerous structures
- Provide information to rescue teams
- Enter areas too dangerous for humans
- NIST -> develops and tests emergency response robot standards
- Focus -> mobility, navigation, sensing and reliability

The challenges and the future

- Unpredictable weather conditions
- Sensor failures
- GPS signal loss
- Unexpected obstacles
- Understanding human behaviour
- Making safe decisions
- Reliability -> especially important in medical deliveries and rescue operations
- Future -> better AI, machine learning and sensors
- More autonomous, adaptable and capable robots
- Safety and reliability remain essential

5. References

1. MDPI - https://www.mdpi.com/2076-3417/13/17/9877

2. Waymo. How Our Cars Drive - https://support.google.com/waymo/answer/9190838

3. Waymo. The Waymo Driver - https://waymo.com/waymo-driver/

4. EASA. Understanding European Drone Regulations and Uses for Drones - https://www.easa.europa.eu/en/domains/drones-air-mobility/drones-air-mobility-landscape/Understanding-European-Drone-Regulations-and-the-Aviation-Regulatory-System

5. Starship Technologies. Autonomous Delivery Robots - https://www.starship.xyz/

6. Gavi - https://www.gavi.org/news/media-room/rwanda-launches-worlds-first-national-drone-delivery-service-powered-zipline

7. NIST. Performance of Emergency Response Robots - https://www.nist.gov/programs-projects/performance-emergency-response-robots

8. NIST. Department of Homeland Security Response Robot Performance Standards — https://www.nist.gov/programs-projects/department-homeland-security-response-robot-performance-standards